

Problem 7

Problem 7

Problem 1 Compute the integral: $\int \frac{1+3x}{4+4x^2} dx$.

Explanation. First notice that

$$\int \frac{1+3x}{4+4x^2} dx = \frac{1}{4} \int \frac{1+3x}{1+x^2} dx = \frac{1}{4} \int \left(\frac{1}{1+x^2} + \frac{3x}{1+x^2} \right) dx.$$

The first integral is $\arctan(x)$, so we have

$$\int \frac{1+3x}{4+4x^2} dx = \frac{1}{4} \arctan(x) + \frac{3}{4} \int \frac{x}{1+x^2} dx.$$

We can evaluate the second integral by substitution. If we set $u = 1+x^2$, then $du = 2x dx$ and $\frac{1}{2} du = x dx$.

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln (1+x^2) + C.$$

Plugging this into our calculation from above gives

$$\int \frac{1+3x}{4+4x^2} dx = \frac{1}{4} \arctan(x) + \frac{3}{8} \ln (1+x^2) + C.$$